

Aiden Luo

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EDUCATION

University of California Santa Cruz

Expected Grad: June 2028

B.S. Robotics Engineering

GPA : 3.78/4

- Coursework: Computer Systems/Assembly Language, Programming Abstractions in Python, Robot Automation, SOE Calculus III, Ordinary Differential Equations, Logic Design, Applied Discrete Math, Intro Probability Theory

EXPERIENCE

IPUSMOBOT (Jiazhi Technology)

Hangzhou, China

Robotics Algorithm Intern

Beginning July 2026

- Incoming algorithm intern contributing to motion planning, sensor fusion, and autonomous navigation
- Applying control systems, sensor integration, and Python/C++ programming to real-world robotics development

Math Tutor

Feb. 2025 – Present

Preply

Online

- Provide personalized one-on-one math tutoring to 40+ students ranging from elementary to college level
- Maintain a 5-star tutor rating and “Super Tutor” status on the platform with over 500 lessons taught

Lego Engineer Instructor

Sept. 2023 – Feb. 2024

Level Up Enrichment

San Diego, CA

- Led after-school STEM LEGO engineering classes for elementary students, teaching concepts like structural design
- Encouraged teamwork through group challenges and maintained Lego kits for smooth operations

PROJECTS

Fog Collector Door Mechanism | CAD, Drivetrain Design, 3D Printing

Professor-Sponsored Research

- Redesigned a 4-motor door system to a single-motor chain-driven system, cutting component costs by 25%
- Designed the mechanism, 3D printed a 20% scale prototype, and designed the full-scale version for outdoor use

Sentinel AI | Python, LLM Agents, OpenRouter, OpenClaw

NVIDIA Hackathon

- Built a multi-agentic AI system that simulates, detects, and autonomously investigates cloud security breaches
- Architected 6 specialized LLM agents using NVIDIA Nemotron via OpenRouter and the OpenClaw agent

Pan-Tilt Tracking Turret | Teensy, OpenCV, C++, Computer Vision

- Built a tracking turret using a Teensy microcontroller, camera module, and dual motors for target following
- Implemented real-time object detection and tracking with OpenCV to calculate target position from live video

AWARDS & ACTIVITIES

Eagle Scout | Senior Patrol Leader

Jan. 2020 – Oct. 2024

Boy Scouts of America

San Diego, CA

- Led and mentored 80+ Scouts as Senior Patrol Leader, overseeing troop activities and leadership development
- Held roles as Patrol Leader, Fire/First Aid/Cooking/Knife Instructor, and Librarian

Presidential Volunteer Service Award

2020 – 2023

Youth Development Resource Center (YDRC) & Chinese Association of Science and Technology

Online

- Led online classes in English, magic, and origami for 100+ students through YDRC and SD-CAST
- Recognized with Bronze (2020), Gold (2021 & 2022), and Silver (2023) Presidential Volunteer Service Awards

TECHNICAL SKILLS

Languages: Python, C/C++, Java

Tools & Frameworks: ROS 2, MuJoCo, OpenCV, Docker, Git, Supabase

Hardware: Teensy, Motors, CAD (Fusion 360 and Onshape), 3D Printing, Raspberry Pi, Arduino Nano, ESP32

Languages Spoken: English (Fluent), Cantonese (Conversational), Mandarin (Conversational)

Interests: Hiking, Camping, Backpacking, Surfing, Fishing, Martial Arts (12 years), Magic Tricks